

Research Article

A Dermatoglyphic Study of Primary Fingerprints Pattern in Relation to Gender and Blood Group Among Residents of Kathmandu Valley, Nepal

Sushma Paudel , Sushmita Paudel , and Samikshya Kafle 

Department of Biotechnology, Himalayan WhiteHouse International College, Purbanchal University, Kathmandu, Bagmati Province, Nepal

Correspondence should be addressed to Sushma Paudel; btec18_sushmapoudel@whitehouseecmt.edu.np

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Fingerprints are unique biometric identifiers that reflect intricate genetic and environmental/physiological influences. Beyond their forensic significance, they can offer insights into physiological traits like blood groups and gender, which can help in forensic analysis to narrow down the search. This exploratory study aims to identify potential associations between fingerprint patterns, gender, and blood groups within a defined regional cohort in Kathmandu, Nepal. This preliminary study included 290 students (144 males and 146 females) from Himalayan Whitehouse International College (HWIC). Fingerprint patterns (loops, whorls, and arches) were analyzed and compared with participants' ABO-Rh blood groups. Statistical analyses, including chi-square tests, were used to determine associations and trends. Loops emerged as the most common fingerprint pattern (57.14%), followed by whorls (35%), and arches (7.86%). Blood group B+ve was the most prevalent (33.1%) among the study population in Kathmandu. The significant association between gender and fingerprint pattern was observed. The gender analysis revealed that loops were predominant in females, while males showed a higher frequency of whorls. While no significant relationship was observed between ABO blood groups and fingerprint patterns, a strong association was found between fingerprint patterns and Rh factor ($p = 0.0496$). Loops were more prevalent among Rh-positive (Rh+ve) individuals, while whorls were more common among Rh-negative (Rh-ve) individuals. Additionally, specific fingers were observed to have distinct fingerprint patterns more frequently. Arches were most prevalent in the index finger of both hands, loops were most abundant in both pinky finger, and left middle finger. Whorls were most frequently observed in ring finger of both hands and right thumb. The findings reinforce global patterns of blood group and fingerprint distribution, where Rh+ve individuals represent the majority and loops are most dominant fingerprint pattern. The gender-specific trends suggest the nuanced interplay of genetics, with females displaying a higher frequency of loops and males showing more whorls. Similarly, some blood group are more likely to exhibit a specific set of fingerprint patterns. This research clearly shows the gender-based differences and influence of genetic factors on fingerprint patterns, particularly the Rh factor. These findings contribute to the growing field of dermatoglyphics, with implications for forensic science, and population genetics.

Keywords: biometric; blood group; dermatoglyphics; fingerprint

1. Introduction

The term “dermatoglyphics” was coined by Anatomist Harold Cummins in 1926. Dermatoglyphics is defined as the scientific study of epidermal markings like ridges and their arrangements; flexion creases and other secondary folds present on

digits, palm of hands, sole or tails with similar skin types [1]. The crucial event of formation of primary ridges patterns on skin's basement membrane starts at the 10th week of gestation and they take obvious visible structural form during 12th–16th week and completed around 24th week of intrauterine embryonic development [2, 3]. These morphological designs are

influenced by genetics and some ecological factors that can apply stress and tension [1]. The fingerprint patterns are very unique to an individual, that there are no two exactly same fingerprints. The probability of occurrence of two people with same fingerprint is extremely low such that, it will occur one in 64 thousand million. Additionally, they rarely change throughout lifetime [2]. Hence, fingerprint patterns are one of the most acceptable and reliable forensic identification tools used as crucial piece of evidence because they are individualistic, constant, universal, inimitable, classifiable, and ubiquitous in any criminal or noncriminal scenes [4, 5]. In the criminal setting, commonly, the fingerprints are used to draw the connection between suspect or criminal to the crime scene or to the subjects related to crime. While in noncriminal civic matter, they are used as biometrics and during the time of disasters for the identification purpose [5]. In addition to fingerprints, lip prints, footprints, bite mark, DNA profiles, and iris scans can be used for identification purpose [2, 6]. In forensic, experts apply numerous classification systems, like the Henry classification or automated fingerprint identification systems (AFIS), to accurately classify and compare fingerprints [7]. There are three primary types of fingerprint patterns: loop pattern, whorl pattern, and arch pattern. Loop has one delta, whorl has two, and arch has one central or no delta at all, and there are radial loop or ulnar loop based on the side to which loop bends [8]. Blood groups are classified on the basis of antigens presence or absence on the surface of red blood cells (RBC). In "ABO" system, the blood is classified as A, B, AB, and O blood group types according to presence or absence of A and/or B antigen. Further, "Rhesus" system classifies blood into rhesus positive (Rh+ve) and rhesus negative (Rh-ve) on the basis of presence or absence of "D" antigen. The ABO blood group system was discovered in 1901 A.D. by Karl Landsteiner at the University of Vienna. ABO and rhesus systems are used in combination in clinical practice [6, 9]. Additionally, gender and blood group are also important pieces of information helpful for shortlisting the suspects or victim while solving crimes and for identification purpose. Interestingly, some forensic research has been done to explore the potential correlation among biometric trait like fingerprint pattern with demographic factors, like gender and blood group. It is being introspected that we can determine sex and blood group from available data of fingerprint from the scene to help identify individuals with ease [4]. Some prevalent studies suggest the differences in fingerprints for their ridges density, pattern types, and dimensions between males and females. It can suggest the potential influence of hormones during development or genetic predisposition for their formation. This highlights the potential of utilizing the fingerprint as biomarkers for gender determination. Furthermore, there has been emerging studies that suggest a potential association between certain fingerprint patterns and specific blood groups [7]; having deeper understanding of these association would help forensic analysis and investigations by aiding in the process of identification. This exploratory study is designed within a defined regional cohort with an effort to advance our understanding of the complex relationship between fingerprint pattern and physiological characteristics, like gender and blood group in forensic science. By examining these relations,

researchers can further utilize the the trends observed and the biometric traits in foundation for further hypothesis-driven research, identification process, and interpreting the criminal incidents.

2. Methods and Methodology

This preliminary study was carried out in 2023 on 290 adults (144 Males and 146 Females) including students, teacher and staffs of HWIC, Kathmandu, Nepal (shown in Figure 1). Only the volunteers with the blood group identified from authorized health care bodies were selected and their fingerprints were studied. Also, the individuals with any hand deformity and permanent disfigurement on one or more fingers were excluded from the study. To the selected subjects, the objective of study and procedure of fingerprint collection were explained in detail and the informed consent form was signed. A fingerprint impression was taken on a white piece of paper with two rows, marked as right and left, and each divided into five columns (namely thumb, index, middle, ring, and pinky as shown in Figure 2). Using the black rubber stamp pads, prints of all ten fingers of both hands were transferred in the respective cells in the paper. This Ink method was demonstrated by Cummins and Mildo in 1961 [10]. Care was taken to avoid smudging of the print.

Name, sex, age, ABO, and Rh blood group were noted for studying the relation between fingerprints pattern and blood group. Then, with the help of a magnifying lens, the primary fingerprint patterns were analyzed and classified as loop, whorls, or arches, based on the appearance of lines using Michael Kuchen's fingerprint classification system [11].

The distribution of dermatoglyphic fingertip patterns in both hands of individuals and its observed associations with demographic traits like gender and different ABO and Rh blood groups were evaluated and analyzed statistically using Excel spreadsheets and R version 4.0.4. Chi square (χ^2) test or Fisher exact tests were used to compare variables and tests were considered significant when p -value was less than 0.05.

3. Result

The present study was based on a sample of 290 individuals only, which, while adequate for preliminary exploration, may limit the generalization of the results, especially across larger or genetically diverse population in forthcoming statistical observations.

3.1. Demographic Distribution in Sample. Out of the 290 study participants, males (144, 49.66%) and females (146, 50.34%) (Shown in Table 1) were nearly equal in proportion.

3.2. Distribution of Blood Groups in the Population and Among Different Genders. The most common blood group among the participants was B+ve (33.1%), followed by A+ve (26.21%) and O+ve (25.52%), with B-ve (1.03%) being the least common (refer to Table 2). The AB-ve blood group (0.0%) was absent in our participants pool. In terms of the Rh factor classification, approximately 94.14% of participants were Rh+ve, while the remaining were Rh-ve. Among Rh-ve, O-ve had the most of the cases (3.10%). In both males and females, B+ve is of high occurrence, followed by A+ve being

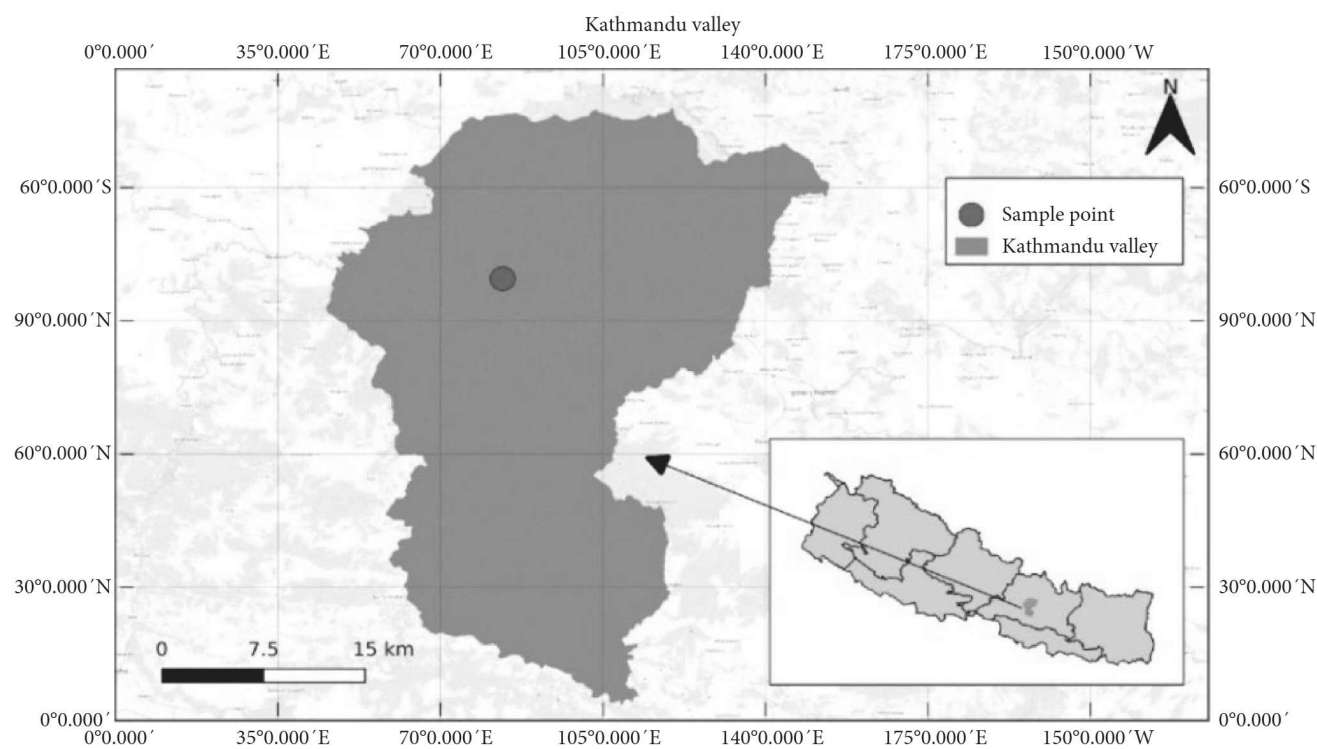


FIGURE 1: Study area in Kathmandu valley, Nepal.

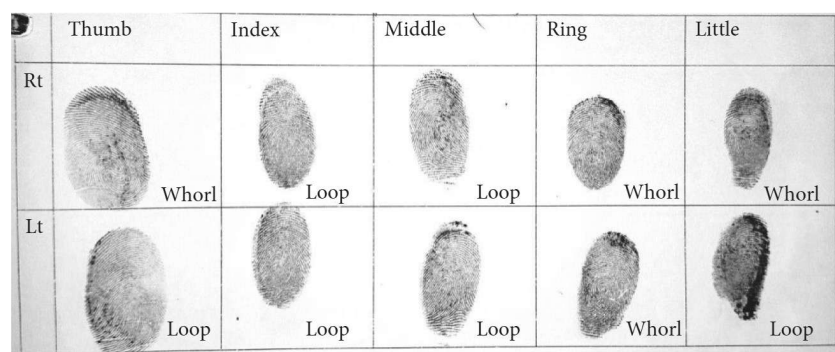


FIGURE 2: Fingerprint impression collected during study classified into loop, whorl, and arch for further analysis.

TABLE 1: Frequency of gender in study sample.

Gender composition	N	N (%)
Female	146	50.34
Male	144	49.66
Grand total	290	100.00

second most high occurrence in females and O+ve being that of males. However, the observed differences in blood group distribution between genders were not statistically significant ($p > 0.05$), indicating that blood group distribution is not determined by sex in this sample.

3.3. Fingerprint Patterns Among Different Genders. Analyzing the primary fingerprint patterns across all fingers on both hands, the most prevalent fingerprint pattern among the

participants, overall, was loops (57.14%) followed by whorls (35.0%), and arches (7.86%). While comparing the fingerprint pattern between male and female participants, loops were the most common pattern in both genders (56.18% in males versus 58.08% in females), followed by whorls (37.92% in males versus 32.12% in females). There was no significant statistical variation between fingerprint pattern for the genders ($p = 0.2281e-05$), that is the fingerprint pattern is associated with gender (Table 3).

3.4. Fingerprint Patterns in Different ABO and Rh Blood Groups. The distribution of fingerprint patterns by ABO group is displayed in Table 4. The blood group A had the highest occurrence (%) of loop, followed by B, AB, and O. Conversely, blood group O exhibited the highest percentage of whorls, followed by AB, B, and A. It was observed that there was a statistically significant difference in fingerprint patterns among

TABLE 2: Distribution of participants according to blood group and gender ($N = 290$).

Blood group	Female		Male		Total	
	<i>N</i>	(%)	<i>N</i>	(%)	<i>N</i>	(%)
A−ve	1	0.68	4	2.78	5	1.72
A+ve	41	28.08	35	24.31	76	26.21
AB+ve	13	8.90	14	9.72	27	9.31
B−ve	1	0.68	2	1.39	3	1.03
B+ve	46	31.51	50	34.72	96	33.10
O−ve	6	4.11	3	2.08	9	3.10
O+ve	38	26.03	36	25.00	74	25.52
Grand total	146	100.00	144	100.00	290	100.00

Note: $\chi^2 = 3.8512$, $df = 6$, p -value = 0.6968, $p > 0.05$; NS.

TABLE 3: Distributions of primary fingerprint patterns in all fingers of both hands by gender ($N = 2900$).

Patterns	Female		Male		Total	
	<i>N</i>	(%)	<i>N</i>	(%)	<i>N</i>	(%)
Arch	143	9.79	85	5.90	228	7.86
Loop	848	58.08	809	56.18	1657	57.14
Whorl	469	32.12	546	37.92	1015	35.00
Grand total	1460	100.00	1440	100.00	2900	100.00

Note: $\chi^2 = 21.377$, $df = 2$, p -value = 2.281e−05, $p < 0.05$.

TABLE 4: Distribution of primary fingerprints patterns in all fingers by ABO blood group system ($n = 2900$).

Patterns	A		B		AB		O	
	<i>N</i>	(%)	<i>N</i>	(%)	<i>N</i>	(%)	<i>N</i>	(%)
Arch	57	7.04	80	8.08	14	5.19	77	9.28
Loop	483	59.63	566	57.17	154	57.04	454	54.70
Whorl	270	33.33	344	34.75	102	37.78	299	36.02
Total	810	100	990	100	270	100	830	100

Note: $\chi^2 = 8.5861$, $df = 6$, p -value = 0.1982; $p > 0.05$; NS.

TABLE 5: Distribution of fingerprints pattern in all fingers by Rh blood group ($n = 2900$).

Print	Rh−ve		Rh+ve		Total	
	<i>N</i>	(%)	<i>N</i>	(%)	<i>N</i>	(%)
Arch	12	7.06	216	7.91	228	7.86
Loop	73	42.94	1584	58.02	1657	57.14
Whorl	85	50.00	930	34.07	1015	35.00
Grand total	170	100	2730	100	2900	100

the ABO blood group ($p = 0.1982$), indicating no association between ABO system with fingerprint pattern distribution (Table 4).

The distribution of fingerprint patterns among Rh +ve and Rh −ve is shown in Table 5. Loops were more common among Rh +ve individuals (58.02%) compared to the Rh −ve individuals (42.94%), while whorls were more common among Rh −ve individuals (50.0%) compared to Rh +ve individuals

(34.07%). Since the data was imbalanced due to significantly higher number of Rh +ve, Generalized Linear Model (GLM) was used with a Poisson distribution to examine the observed association between these variables.

The results indicate that there is a observed association between Rh type and fingerprint pattern distribution, with a significant difference found between the two groups (p -value = 0.0496). This suggests that blood group, particularly the Rh factor, is associated with differences in fingerprint pattern frequency. Overall, arches were most prevalent among A−ve (12%) and O+ve (9.59%), and B+ve (8.33%) blood groups. Loops were most common in A+ve (60.13%), followed by B+ve (57.29%), O+ve (57.16%), and AB+ve (57.04%). Whorls were most common in O−ve (58.89%), followed by B−ve (46.67%) (refer to Table 6).

3.5. Fingerprint Pattern Among Different Blood Group of Males And Females. The distribution of fingerprint patterns across different ABO and Rh blood groups in males and females showed a statistically significant association, indicated by a p -value less than 0.05. (Table 7)

3.6. Fingerprint Pattern in Different Fingers. Table 8 shows that arches were most prevalent in the right index finger (23.25%), followed by the left index finger (20.61%). Loops were most abundant in the left pinky finger (12.73%), right pinky finger (12.61%), and left middle finger (11.16%). Whorls were most frequently observed in the right ring finger (15.47%), left ring finger (14.38%), and right thumb (12.51%).

4. Discussion

Fingerprints are impressions made by the fine ridges, present at the fingertips, and are highly individualistic. The fingerprints are of key importance in establishing the identity of the culprits who were present at the scene of the crime along with the victims of mass disasters [12]. This exploratory study shows the relationship between gender, ABO, Rh blood groups, and dermatoglyph distribution (dactylography and fingerprint). There were 290 individuals in all, 144 male and 146 female, selected at random from Himalyan Whitehouse International College, Kathmandu. In study sample, the observed sex distribution (50.34% female and 49.66% male) is same as the expected distribution based on Nepal's 2021 census data (51.1% female and 48.9% male) [13]. The distribution of ABO and Rh blood groups vary from population to population. The current study showed that in Kathmandu city, Rh+ve have the highest number of individuals. The general distribution of blood groups was of the same order with that of the present study population that is highest frequency of Rh+ve and lowest frequency of Rh−ve. [14, 15]. Highest frequency of blood group B, followed by O, then A, and lowest blood group AB was observed (Look at Figure 3). Blood group B+ve having 33.1% in the population is the highest occurring in the study sample. These results differed from distribution among the Nepalese population shown by another study done in Chitwan with order O (36.4%), B (30.7%), A (25.7%), and AB (7.2%) [9]. Again, studies suggested that the most common one is O+ve and least common is AB−ve [9, 14, 15]. Recent findings cleared

TABLE 6: Distribution of primary fingerprint patterns in all fingers of both hands by ABO and Rh blood groups ($n=2900$).

Blood group		Arch	Loop	Whorl	Grand total
A	Rh−ve	6 (12%)	26 (52%)	18 (36%)	50 (100%)
	Rh+ve	51 (6.71%)	457 (60.13%)	252 (33.15%)	760 (100%)
B	Rh−ve	0 (0%)	16 (53.33%)	14 (46.67%)	30 (100%)
	Rh+ve	80 (8.33%)	550 (57.29%)	330 (34.37%)	960 (100%)
AB	Rh−ve	0	0	0	0
	Rh+ve	14 (5.19%)	154 (57.04%)	102 (37.78%)	270 (100%)
O	Rh−ve	6 (6.67%)	31 (34.44%)	53 (58.89%)	90 (100%)
	Rh+ve	71 (9.59%)	423 (57.16%)	246 (33.24%)	740 (100%)

Note: $\chi^2 = 142.18$, $df = 30$, p -value $< 2.2e-16$; $p < 0.05$.

TABLE 7: Distribution of primary fingerprint pattern among different gender with different blood group.

Blood	Arch			Loop			Whorl		
	F	M	Total	F	M	Total	F	M	Total
A−ve	0	6	6	3	23	26	7	11	18
A+ve	30	21	51	267	190	457	113	139	252
AB+ve	5	9	14	77	77	154	48	54	102
B−ve	0	0	0	10	6	16	0	14	14
B+ve	52	28	80	273	277	550	135	195	330
O−ve	6	0	6	19	12	31	35	18	53
O+ve	50	21	71	199	224	423	131	115	246
Total	143	85	228	848	809	1657	469	546	1015

Note: $\chi^2 = 142.18$, $df = 30$, p -value $< 2.2e-16$; $p < 0.05$.

TABLE 8: Distribution of primary fingerprint pattern in different fingers ($n=2900$).

Finger	Arch		Loop		Whorl	
	N	(%)	N	(%)	N	(%)
Left_index	47	20.61	142	8.57	101	9.95
Left_middle	28	12.28	185	11.16	77	7.59
Left_pinky	12	5.26	211	12.73	67	6.60
Left_ring	14	6.14	130	7.85	146	14.38
Left_thumb	19	8.33	172	10.38	99	9.75
Right_index	53	23.25	134	8.09	103	10.15
Right_middle	25	10.96	198	11.95	67	6.60
Right_pinky	10	4.39	209	12.61	71	7.00
Right_ring	12	5.26	121	7.30	157	15.47
Right_thumb	8	3.51	155	9.35	127	12.51
Total	228	100	1657	100	1015	100

the most frequent blood group is B+ve in both females (31.51%) and males (33.1%), but [15] suggested the most common O+ve blood type in males or A+ve in females in Nepal, which is similar to the case of the second most common blood group type among males and females in the present study.

In the distribution of fingerprint pattern within Kathmandu, the loop was the most frequent fingerprint type and arch was the least common. This distribution is consistent with studies conducted in various populations, which have similarly reported loops as the dominant pattern and arches

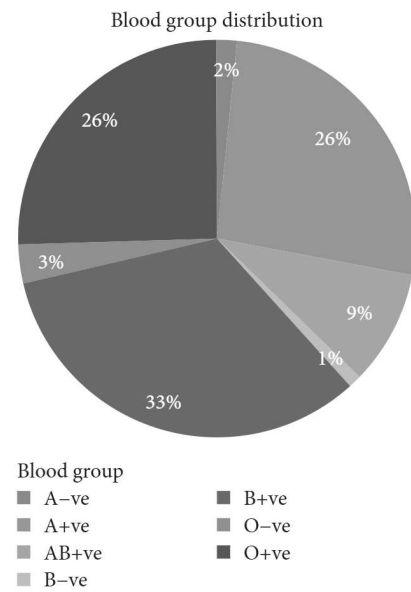


FIGURE 3: Blood group distribution among the sample population.

as the least prevalent [3, 5, 6, 8]. Additionally, our study found significant relationship between fingerprint patterns and gender, further corroborating findings from [16–18]. These consistent observations across diverse populations strengthen the hypothesis that genetic factors predominantly influence fingerprint pattern distribution, with significant or minimal impact from gender. In observation of the fingerprint pattern distribution in our study, arches and loops patterns were found to be in higher frequencies in females which is congruent to the studies by [17, 19] among Nepalese Community and Oman Community, respectively. However, some studies have reported the higher occurrence of loops in males and whorls in females [6]. The current study shows no significant association between the ABO blood group and fingerprint pattern while there is the significant association between the fingerprint and Rh blood types. On analysis of relation among combined blood group type and fingerprint, the significant association was observed. This is in congruent with the blood group A had the highest occurrence (%) of loop, followed by B, AB, and O. Conversely, blood group O exhibited the highest percentage of whorls, followed by AB, B, and A. Loops were more common among Rh+ve individuals (58.02%) compared to the Rh−ve

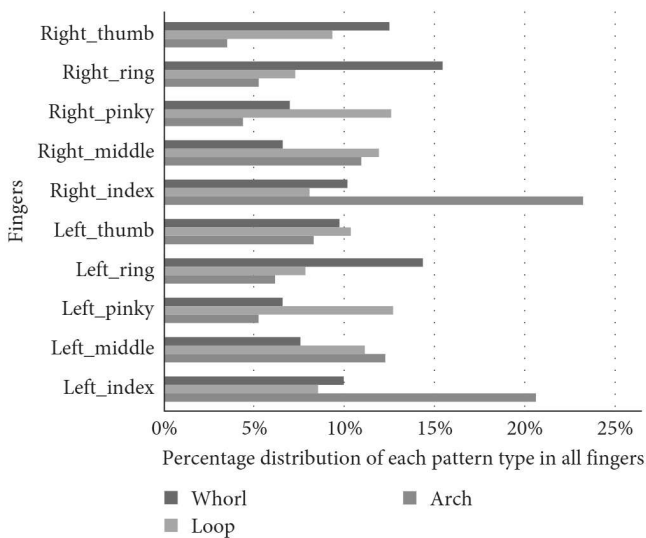


FIGURE 4: Graph showing percentage distribution of each fingerprint pattern in different blood group.

individuals (42.94%), while whorls were more common among Rh-ve individuals (50.0%) compared to Rh+ve individuals (34.07%). Overall, arches were most prevalent among A-ve (12%), O+ve (9.59%), and B+ve (8.33%) blood groups among the population of Kathmandu. The study by Gowri S R and G [2] presented A-ve (20%), AB-ve (20%), B-ve (20%), and O-ve (20%) with higher prevalence and another study [19] showed arches most prevalent among AB+ve (7.37%), O-ve (7%), and B+ve (6.8%). Loops were most common in A+ve (60.13%), followed by B+ve (57.29%), O+ve (57.16%), and AB+ve (57.04%) but [2] reported AB+ve (70.83%), A-ve (65%), O-ve (56.25%), and O+ve (54%) in decreasing order. Another study by KC et al. [19] showed O+ve (53%), A+ve (52%), and B+ve (51%) order for the loop prevalence. Present research establish whorls to be most common in O-ve (58.89%), followed by B-ve (46.67%) which doesn't match with the report from A+ve (32.3%) and B+ve (31.9%) shown by Gowri S R and G [2]. While [19] established, blood group A-ve (74%) to have most whorls occurrence followed by B-ve (56%). Arches were most prevalent in the right index finger (23.25%) followed by the left index finger (20.61%). Loops were most abundant in the left pinky finger (12.73%), right pinky finger (12.61%), and left middle finger (11.16%). Whorls were most frequently observed in the right ring finger (15.47%), left ring finger (14.38%), and right thumb (12.51%) (as in Figure 4). Study in Kathmandu by KC et al. [19] showed ring finger with the high frequency of whorls in all blood groups which corroborates to our finding.

This study cleared the relationship of three variables. The findings of the study can be concluded as the loops are the most commonly occurring fingerprint pattern and arches are the least common, blood group B+ve is the most common, and loops are predominant in blood group A, B, AB, and O in both Rh+ve and Rh-ve individuals, except in O-ve where whorls are more common and males have a higher incidence of whorls and females have a higher incidence of loops.

5. Conclusion

In conclusion, loops are the most common fingerprint pattern observed among both males and females within the population in Kathmandu Valley, Nepal. It is followed by whorls, and arches are least common. This study concludes that the distribution of primary fingerprint patterns correlates with the ABO-Rh conventional blood group types and gender but not solely with ABO factors. Additionally, the specific fingers tend to exhibit distinct fingerprint patterns more frequently.

The findings suggest that fingerprint patterns can be used to predict an individual's gender and ABO-Rh blood group and vice versa, which could increase the chances of identifying both victims or criminals. But further study of the population with all blood group inclusion is required to conclude the proper relation. As fingerprints has been the key component of biometric identification and will remain so in the future, the observed relationship between fingerprint patterns and ABO blood groups provides an additional layer of data for personal identification.

5.1. Limitations. Despite the nearly equal distribution of male and female participants, the study was limited by a relatively small sample size and unequal representation of blood groups. In particular, the AB-ve blood group was not included. Further studies with larger, more representative samples from this region are recommended to establish a stronger and clearer relationship between dermatoglyphic patterns and physiological characteristics, such as gender, ABO, and Rh blood groups. This would allow for more accurate predictions and insights into the connection between fingerprint patterns, gender, and blood group classifications. In the study, fingerprints were collected using the traditional ink-on-paper technique, which, while widely used and cost-effective, can result in reduced clarity when scanned or photographed for digital presentation. Future studies should consider using digital fingerprint scanners to enhance image resolution and improve reproducibility in biometric analysis.

Data Availability Statement

Data shall be made available upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

Author Contributions

All authors have read and agreed to the published version of the manuscript.

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